

10

Magnetic trapping, evaporative cooling and Bose–Einstein condensation

10.1 Principle of magnetic trapping	218
10.2 Magnetic trapping	220
10.3 Evaporative cooling	224
10.4 Bose–Einstein condensation	226
10.5 Bose–Einstein condensation in trapped atomic vapours	228
10.6 A Bose–Einstein condensate	234
10.7 Properties of Bose-condensed gases	239
10.8 Conclusions	242
Exercises	243

Magnetic traps are used to confine the low-temperature atoms produced by laser cooling. If the initial atomic density is sufficiently high, the simple but extremely effective technique of evaporative cooling allows experiments to reach quantum degeneracy where the occupation of the quantum states approaches unity. This leads either to Bose–Einstein condensation (BEC) or to Fermi degeneracy, depending on the spin of the atoms. This chapter describes magnetic traps and evaporative cooling, using straightforward electromagnetism and kinetic theory, before giving an outline of some of the exciting new types of experiments that have been made possible by these techniques. The emphasis is on presenting the general principles and illustrating them with some relevant examples rather than attempting to survey the whole field in a qualitative way.

10.1 Principle of magnetic trapping

In their famous experiment, Otto Stern and Walter Gerlach used the force on an atom as it passed through a strong inhomogeneous magnetic field to separate the spin states in a thermal atomic beam. Magnetic trapping uses exactly the same force, but for cold atoms the force produced by a system of magnetic field coils bends the trajectories right around so that low-energy atoms remain within a small region close to the centre of the trap. Thus the principle of magnetic trapping of atoms has been known for many years but it only became widely used after the development of laser cooling.¹

A magnetic dipole μ in a field has energy

$$V = -\mu \cdot \mathbf{B}. \quad (10.1)$$

For an atom in the state $|JFM_F\rangle$ this corresponds to a Zeeman energy

$$V = g_F \mu_B M_F B. \quad (10.2)$$

¹There was early work on magnetic trapping of ultra-cold neutrons whose magnetic moment is only $-1.9 \mu_N$, and the nuclear magneton μ_N , which is much smaller than the Bohr magneton μ_B .

The energy *depends only the magnitude of the field* $B = |\mathbf{B}|$. The energy does not vary with the direction of $\mathbf{B}$ because as the dipole moves (adiabatically) it stays aligned with the field. From this we find the magnetic force along the z -direction:

$$F = -g_F \mu_B M_F \frac{dB}{dz}. \quad (10.3)$$

Example 10.1 *Estimate of the effect of the magnetic force on an atom of mass M that passes through a Stern–Gerlach magnet at speed v .*

The atom takes a time $t = L/v$ to pass through a region of high field gradient of length L (between the pole pieces of the magnet), where it receives an impulse Ft , in the transverse direction, that changes its momentum by $\Delta p = Ft$. An atom with momentum $p = Mv$ along the beam has a deflection angle of

$$\theta = \frac{\Delta p}{p} = \frac{FL}{Mv^2} = \frac{FL}{2E_{\text{ke}}}. \quad (10.4)$$

The kinetic energy $E_{\text{ke}} \simeq 2k_B T$ (from Table 8.1), where T is the temperature of the oven from which the beam effuses. An atom with a single valence electron has a maximum moment of μ_B (when $g_F M_F = 1$) and hence

$$\theta = \frac{\mu_B}{k_B} \times \frac{dB}{dz} \frac{L}{4T} = 0.67 \times \frac{3 \times 0.1}{4 \times 373} \simeq 1.4 \times 10^{-4} \text{ rad}. \quad (10.5)$$

This evaluation for a field gradient of 3 T m^{-1} over $L = 0.1 \text{ m}$ and $T = 373 \text{ K}$ makes use of the ratio of the Bohr magneton to the Boltzmann constant, given by

$$\frac{\mu_B}{k_B} = 0.67 \text{ K T}^{-1}. \quad (10.6)$$

Thus a well-collimated beam of spin-1/2 atoms that propagates for $L = 1 \text{ m}$ after the magnet will be split into two components separated by $2\theta L = 0.3 \text{ mm}$.

For an atom with $T \simeq 0.1 \text{ K}$ eqn 10.5 gives a deflection of 0.5 rad ! Although the equation is not valid for this large angle, it does indicate that magnetic forces have a strong influence on laser-cooled atoms and can bend their trajectories around. From eqn 10.6 we see directly that a magnetic trap where the field varies from 0 to 0.03 T has a depth of 0.02 K , e.g. a trap with a field gradient of 3 T m^{-1} over 10 mm , as described in the next section. Remember that the Doppler cooling limit for sodium is $240 \mu\text{K}$, so it is easy to capture atoms that have been laser cooled. Traps made with superconducting magnetic coils can have fields of over 10 T and therefore have depths of several kelvin; this enables researchers to trap species such as molecules that cannot be laser cooled.²

²Standard laser cooling does not work for molecules because repeated spontaneous emission causes the population to spread out over many vibrational and rotational levels. Superconducting traps operate at low temperatures with liquid helium cooling, or a dilution refrigerator, and molecules are cooled to the same temperature as the surroundings by buffer gas cooling with a low pressure of helium (cf. Section 12.4).

10.2 Magnetic trapping

10.2.1 Confinement in the radial direction

The estimate in the introduction shows that magnetic forces have a significant effect on cold atoms and in this section we examine the specific configuration used to trap atoms shown in Fig. 10.1. The four parallel wires arranged at the corners of a square produce a quadrupole magnetic field when the currents in adjacent wires flow in opposite directions. Clearly this configuration does not produce a field gradient along the axis (z -direction); therefore from Maxwell's relation $\text{div } \mathbf{B} = 0$ we deduce that

$$\frac{dB_x}{dx} = -\frac{dB_y}{dy} = b'.$$

These gradients have the same magnitude b' , but opposite sign. Therefore the magnetic field has the form

$$\mathbf{B} = b' (x\hat{\mathbf{e}}_x - y\hat{\mathbf{e}}_y) + \mathbf{B}_0. \quad (10.7)$$

Here we simply assume that $b' = 3 \text{ T m}^{-1}$; that this is a realistic field gradient can be shown by using the Biot–Savart law to calculate the field produced by coils carrying reasonable currents (see caption of Fig. 10.1). In the special case of $B_0 = 0$, the field has a magnitude

$$|\mathbf{B}| = b'(x^2 + y^2)^{1/2} = b'r. \quad (10.8)$$

Thus the magnetic energy (eqn 10.2) has a linear dependence on the radial coordinate $r = \sqrt{x^2 + y^2}$. This conical potential has the V-shaped cross-section shown in Fig. 10.2(a), with a force in the radial direction of

$$\mathbf{F} = -\nabla V = -g_F \mu_B M_F b' \hat{\mathbf{e}}_r. \quad (10.9)$$

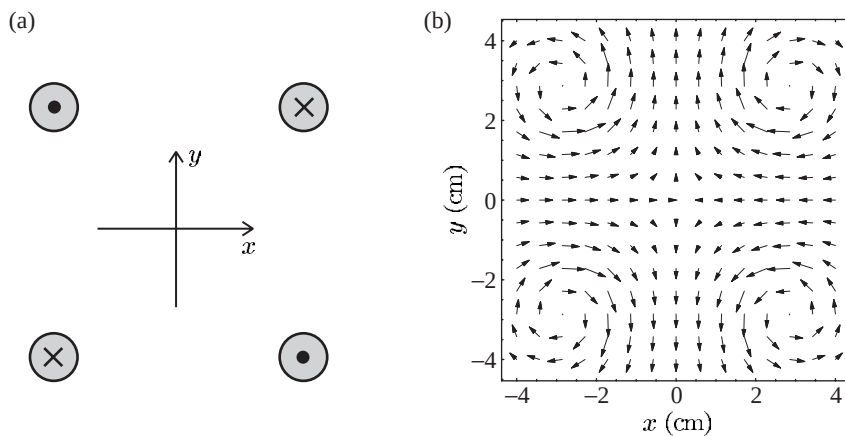


Fig. 10.1 (a) A cross-section through four parallel straight wires, with currents into and out of the page as indicated. These give a magnetic quadrupole field. In a real magnetic trap, each ‘wire’ is generally made up of more than ten strands, each of which may conduct over 100 amps, so that the total current along each of the four wires exceeds 1000 amps. (b) The direction of the magnetic field around the wires—this configuration is a magnetic quadrupole.

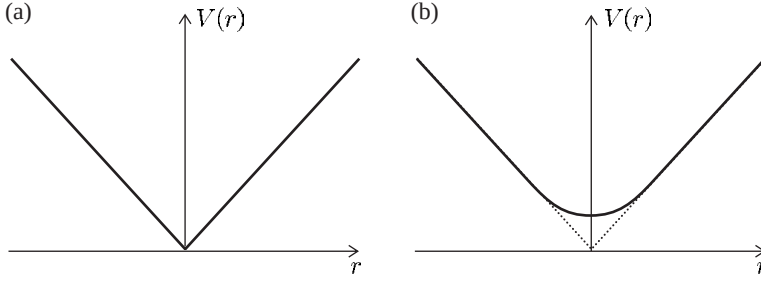


Fig. 10.2 (a) A cross-section through the magnetic potential (eqn 10.8) in a radial direction, e.g. along the x - or y -axis. The cusp at the bottom of the conical potential leads to non-adiabatic transitions of the trapped atoms. (b) A bias field along the z -direction rounds the bottom of the trap to give a harmonic potential near the axis (in the region where the radial field is smaller than the axial bias field).

This force confines atoms in a low-field-seeking state, i.e. one with $g_F M_F > 0$, so that the magnetic energy decreases as the atom moves into a lower field (see Fig. 6.10, for example). However, a quadrupole field has a fundamental problem—the atoms congregate near the centre where $B = 0$ and the Zeeman sub-levels ($|JFM_F\rangle$ states) have a very small energy separation. In this region of very low magnetic field the states with different magnetic quantum numbers mix together and atoms can transfer from one value of M_F to another (e.g. because of perturbations caused by noise or fluctuation in the field). These non-adiabatic transitions allow the atoms to escape and reduce the lifetime of atoms in the trap. The behaviour of atoms in this magnetic trap with a leak at the bottom resembles that of a conical funnel filled with water—a large volume of fluid takes a considerable time to pass through a funnel with a small outlet at its apex, but clearly it is desirable to plug the leak at the bottom of the trap.³

The loss by non-adiabatic transitions cannot be prevented by the addition of a uniform field in the x - or y -directions, since this simply displaces the line where $\mathbf{B} = \mathbf{0}$, to give the same situation as described above (at a different location). A field $\mathbf{B}_0 = B_0 \hat{\mathbf{e}}_z$ along the z -axis, however, has the desired effect and the magnitude of the field in eqn 10.7 becomes

$$|\mathbf{B}| = \{B_0^2 + (b'r)^2\}^{1/2} \simeq B_0 + \frac{b'^2 r^2}{2B_0}. \quad (10.10)$$

This approximation works for small r where $b'r \ll B_0$. The bias field along z rounds the point of the conical potential, as illustrated in Fig. 10.2(b), so that near the axis the atoms of mass M see a harmonic potential. From eqn 10.2 we find

$$V(r) = V_0 + \frac{1}{2} M \omega_r^2 r^2. \quad (10.11)$$

The radial oscillation has an angular frequency given by

$$\omega_r = \sqrt{\frac{g_F \mu_B M_F}{M B_0}} \times b'. \quad (10.12)$$

10.2.2 Confinement in the axial direction

The *Ioffe* trap, shown in Fig. 10.3, uses the combination of a linear magnetic quadrupole and an axial bias field described above to give

³These losses prevent the spherical quadrupole field configuration of the two coils in the MOT being used directly as a magnetic trap—the MOT operates with gradients of 0.1 T m^{-1} so thirty times more current-turns are required in any case. We do not discuss the addition of a time-varying field to this configuration that leads to the TOP trap used in the first experimental observation of Bose–Einstein condensation in the dilute alkali vapours (Anderson *et al.* 1995).

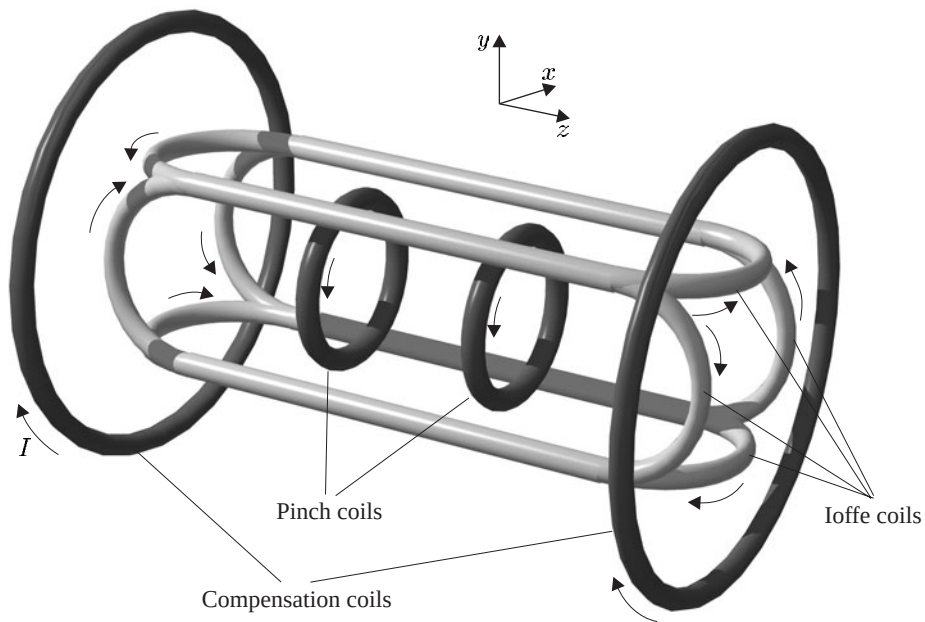


Fig. 10.3 An Ioffe–Pritchard magnetic trap. The fields produced by the various coils are explained in more detail in the text and the following figures. This Ioffe trap is loaded with atoms that have been laser cooled in the way shown in Fig. 10.5. (Figure courtesy of Dr Kai Dieckmann.)

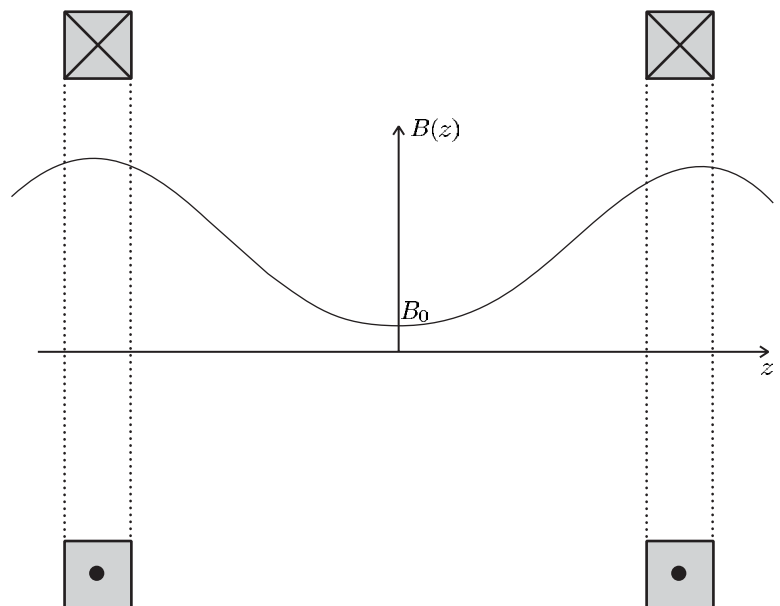


Fig. 10.4 The pinch coils have currents in the same direction and create a magnetic field along the z -axis with a minimum midway between them, at $z = 0$. This leads to a potential well for atoms in low-field-seeking states along this axial direction. By symmetry, these co-axial coils with currents in the same direction give no gradient at $z = 0$.

radial confinement for atoms in low-field-seeking states. To confine these atoms in the axial direction the trap has two pairs of co-axial coils with currents that flow in the same direction to produce a field along the z -axis whose magnitude is shown in Fig. 10.4. These so-called pinch coils⁴ have a separation greater than that of Helmholtz coils, so the field along z has a minimum midway between the coils (where $dB_z/dz = 0$). The

⁴The term ‘pinch coils’ arises from the concept of pinching off the ‘magnetic tube’ containing the atoms. The Ioffe trap configuration was originally developed to confine plasma.

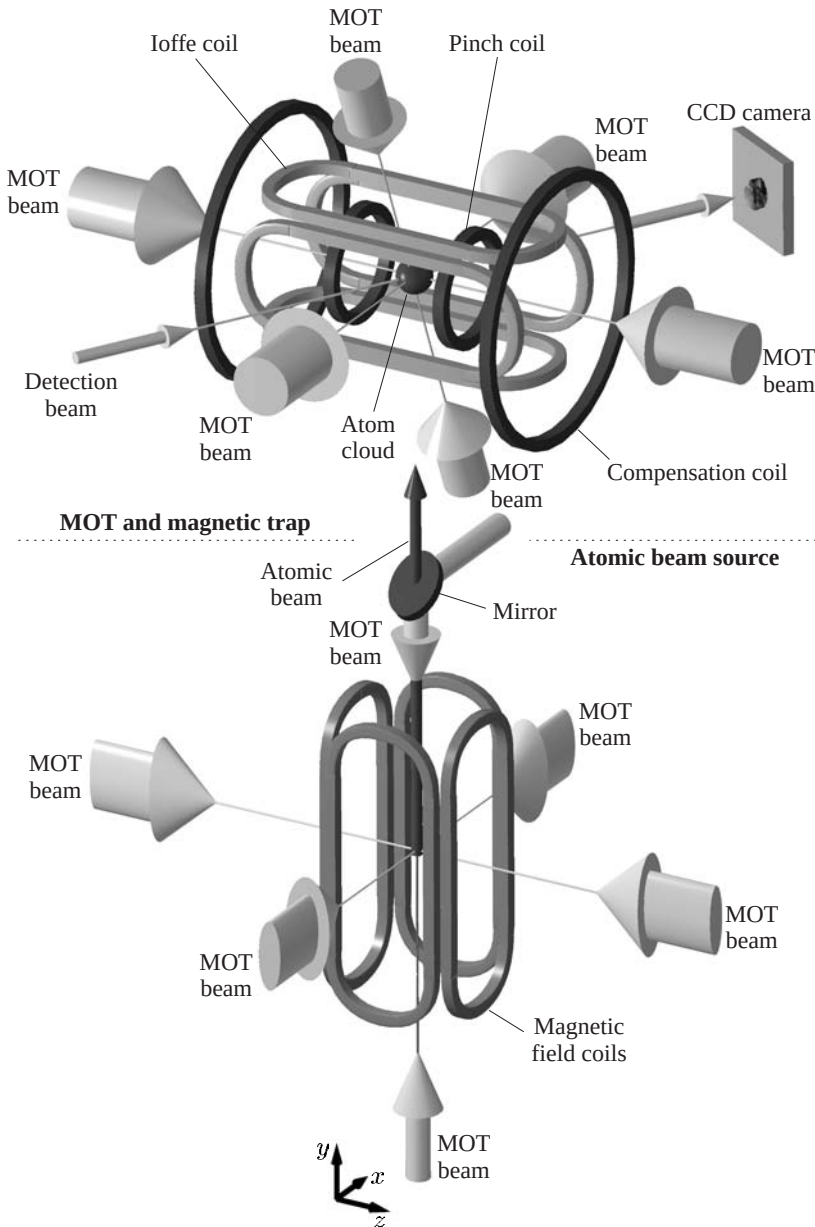


Fig. 10.5 A general view of an apparatus to load an Ioffe–Pritchard magnetic trap with laser-cooled atoms from a magneto-optical trap. (The MOT has a different arrangement of coils to that described in Section 9.4 but the same principle of operation.) This experimental apparatus was constructed by the team of Professor Jook Walraven at the FOM Institute, Amsterdam. (Figure courtesy of Dr Kai Dieckmann, Dieckmann *et al.* (1998).) Copyright 1998 by the American Physical Society.

field has the form

$$B_{\text{pinch}}(z) = B_{\text{pinch}}(0) + \frac{d^2 B_z}{dz^2} \frac{z^2}{2}. \quad (10.13)$$

This gives a corresponding minimum in the magnetic energy and hence a harmonic potential along the z -axis. Typically, the Ioffe trap has an axial oscillation frequency ω_z an order of magnitude lower than ω_r ($= \omega_x = \omega_y$), e.g. $\omega_z/2\pi = 15$ Hz and $\omega_r/2\pi = 250$ Hz (see Exercise 10.1). Thus the atoms congregate in a cigar-shaped cloud along the z -axis. The curvature of the magnetic field along z depends only on the dimensions of the pinch coils and their current. Therefore a uniform field along z does *not* affect ω_z , but it does change ω_r through the dependence on B_0 in eqn 10.12. The pairs of compensation coils shown in Fig. 10.3 create a uniform field along the z -axis that opposes the field from the pinch coils. This allows experimenters to reduce B_0 and make the trap stiff in the radial direction.⁵

To load the approximately spherical cloud of atoms produced by optical molasses, the Ioffe trap is adjusted so that $\omega_r \simeq \omega_z$. After loading, an increase in the radial trapping frequency, by reducing the bias field B_0 (see eqn 10.12), squeezes the cloud into a long, thin cigar shape. This adiabatic compression gives a higher density and hence a faster collision rate for evaporative cooling.

10.3 Evaporative cooling

Laser cooling by the optical molasses technique produces atoms with a temperature below the Doppler limit, but considerably above the recoil limit. These atoms can easily be confined in magnetic traps (as shown in Section 10.1) and evaporative cooling gives a very effective way of reducing the temperature further. In the same way that a cup of tea loses heat as the steam carries energy away, so the cloud of atoms in a magnetic trap cools when the hottest atoms are allowed to escape. Each atom that leaves the trap carries away more than the average amount of energy and so the remaining gas gets colder, as illustrated in Fig. 10.6. A simple model that is useful for understanding this process (and for quantitative calculations in Exercise 10.4) considers evaporation as a sequence of steps. At the start of a step the atoms have a Boltzmann distribution of energies $\mathcal{N}(E) = \mathcal{N}_0 \exp(-E/k_B T_1)$ characteristic of a temperature T_1 . All atoms with energies greater than a certain value $E > E_{\text{cut}}$ are allowed to escape, where $E_{\text{cut}} = \eta k_B T_1$ and typically the parameter η lies in the range $\eta = 3$ –6; this truncated distribution has less energy per atom than before the cut so that, after collisions between the atoms have re-established thermal equilibrium, the new exponential distribution has a lower temperature $T_2 < T_1$.⁶ The next step removes atoms with energies above $\eta k_B T_2$ (a lower energy cut-off than in the first step) to give further cooling, and so on.⁷ For many small steps this model gives a reasonable approximation to real experiments where evaporation proceeds by a continuous ramping down of E_{cut} that cuts away atoms at

⁵In practice, the compensation coils need not have exactly the Helmholtz spacing; the current in these coils together with pinch coil current gives two experimental parameters that allow the adjustment of the magnitude and field curvature along z to any desired value (limited by the maximum current through the pairs of coils). Neither pair of these coils gives a field gradient, by symmetry.

⁶Temperature is only defined at thermal equilibrium and in other situations the mean energy per atom should be used.

⁷An exponential distribution extends to infinity, and so for any value of E_{cut} (or η) there is always some probability of atoms having a higher energy; however, the removal of a very small fraction of atoms has a little effect when averaged over the remaining atoms. Exercise 10.4 compares different depths of cut.

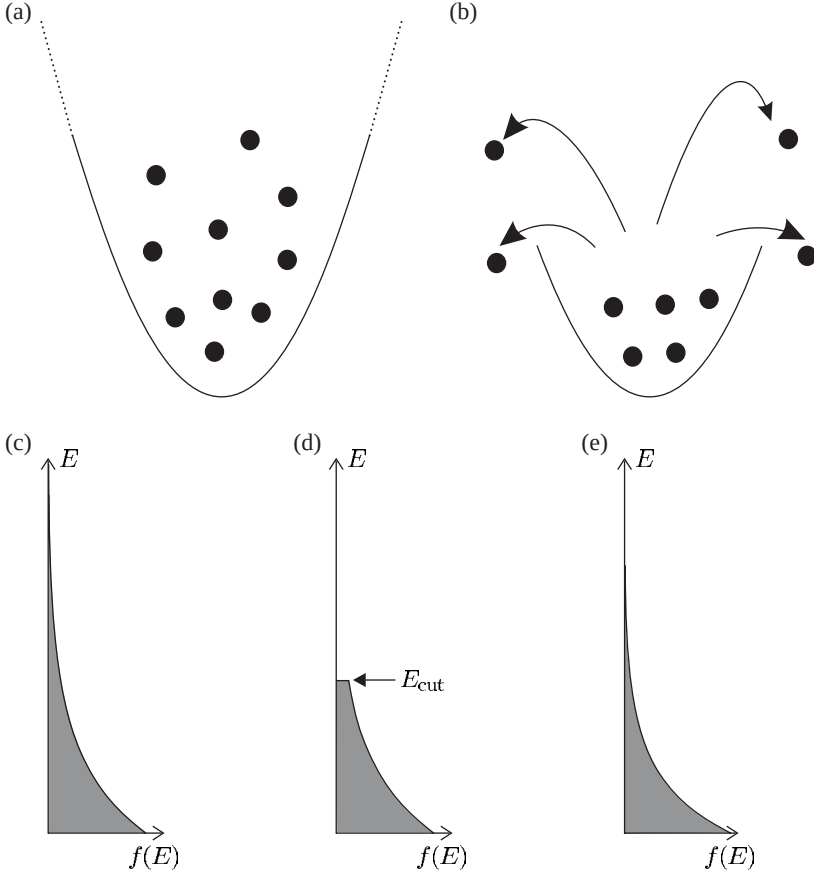


Fig. 10.6 (a) A schematic representation of atoms confined in a harmonic potential. (b) The height of the potential is reduced so that atoms with above-average energy escape; the remaining atoms have a lower mean energy than the initial distribution. The evolution of the energy distribution is shown below: (c) shows the initial Boltzmann distribution $f(E) = \exp(-E/k_B T_1)$; (d) shows the truncated distribution soon after the cut, when the hot atoms have escaped; and (e) shows the situation some time later, after collisions between the remaining atoms have re-established a Boltzmann distribution at a temperature T_2 less than T_1 . In practice, evaporative cooling in magnetic traps differs from this simplified picture in two respects. Firstly, the potential does not change but atoms leave the trap by undergoing radio-frequency transitions to untrapped states at a certain distance from the trap centre (or equivalently at a certain height up the sides of the potential). Secondly, cooling is carried out continuously rather than as a series of discrete steps.

the edge of the cloud (without stopping to allow rethermalisation). The rate of this evaporative cooling ramp depends on the rate of collisions between atoms in the trap.⁸

During evaporation in a harmonic trap the density *increases* (or at least stays constant) because atoms sink lower in the potential as they get colder. This allows *runaway evaporation* that reduces the temperature by many orders of magnitude, and increases the phase-space density to a value at which quantum statistics becomes important.⁹

Evaporation could be carried out by turning down the strength of the trap, but this reduces the density and eventually makes the trap too weak to support the atoms against gravity. (Note, however, that this method has been used successfully for Rb and Cs atoms in dipole-force traps.) In magnetic traps, precisely-controlled evaporation is carried out by using radio-frequency radiation to drive transitions between the trapped and untrapped states, at a given distance from the trap centre, i.e. radiation at frequency ω_{rf} drives the $\Delta M_F = \pm 1$ transitions at a radius r that satisfies $g_F \mu_B b' r = \hbar \omega_{\text{rf}}$. Hot atoms whose oscillations extend beyond this radius are removed, as shown in the following example.

⁸ If the process is carried out too rapidly then the situation becomes similar to that for a non-interacting gas (with no collisions) where cutting away the hot atoms does not produce any more low-energy atoms than there are initially. It just selects the coldest atoms from the others.

⁹ In contrast, for a square-well potential the density and collision rate decrease as atoms are lost, so that evaporation would grind to a halt. In the initial stages of evaporation in an Ioffe trap, the atoms spread up the sides of the potential and experience a linear potential in the radial direction. The linear potential gives a greater increase in density for a given drop in temperature than a harmonic trap, and hence more favourable conditions to start evaporation.

Example 10.2 For an atom with $g_F = 1/2$ (as in sodium) in a trap with $b' = 3 \text{ T m}^{-1}$, the frequency varies with position as $g_F \mu_B b' / h = 21 \text{ GHz m}^{-1}$ ($\mu_B \equiv 14 \text{ GHz T}^{-1}$). The application of radio-frequency radiation at 40 MHz removes atoms over a surface whose cross-section in the plane $z = 0$ is a circle of radius $r = 2 \text{ mm}$. Sweeping the radio-frequency radiation down to 20 MHz reduces the radius to $r = 1 \text{ mm}$. This estimate assumes that the cutting surface (where the radio-frequency radiation removes atoms from the cloud) lies in the region where the magnetic field is linear $b'r \gg B_0$; this is the *opposite* of the condition that gives the harmonic approximation in eqn 10.10. For a bias field of $B_0 = 3 \times 10^{-4} \text{ T}$ and the field gradient b' above, the trapping potential is linear for $r \gg 0.1 \text{ mm}$, so our assumption of a linear field was valid. As evaporation proceeds, the atoms sink further down in the trap and the cross-over from a linear to a harmonic potential occurs when the cloud of atoms has a radius of $r = B_0/b' = 0.1 \text{ mm}$ (see Fig. 10.2(b)). From eqn 10.6 we find that in a linear trap this would correspond to a cloud with a temperature of $2 \times 10^{-4} \text{ K}$.¹⁰

¹⁰This is approximately equal to the Doppler cooling limit for sodium (but see Exercise 10.2(c)).

Evaporative cooling has no fundamental lower limit and temperatures below 10 nK have been reached in magnetic traps. This is sufficient for the experiments discussed here, but let us consider briefly what limitations might arise in practice: (a) for a given set of starting conditions, it is not worthwhile to go beyond the point at which the number of trapped atoms becomes too low to detect;¹¹ (b) when the energy resolution of the radio-frequency transition is similar to the energy of the remaining atoms it is no longer possible to selectively remove hot atoms whilst leaving the cold atoms—colloquially, this is referred to as the radio-frequency ‘knife’ being blunt so that it cannot shave off atoms from the edges of the cloud;¹² and (c) in the case of fermions, it is difficult to cool atoms well below the Fermi temperature T_F at which quantum degeneracy occurs because, when almost all the states with energy below $k_B T_F$ are filled (with the one atom in each state allowed by the Pauli exclusion principle), there is a very low probability of an atom going into an unoccupied state (‘hole’) in a collision. The case of bosons is discussed in the next section.

¹¹Good images were obtained from 2000 rubidium atoms in the first BEC experiment, and in principle it is possible to detect even a single atom.

¹²Contributions to the width of the radio-frequency transitions between Zeeman sub-levels arise from power broadening and fluctuations (noise) in the magnetic field. More usually, radio-frequency spectroscopy has a resolution limited by the interaction time, but this is less important for trapped atoms and continuous radiation.

The temperature of a cloud of trapped atoms can be reduced by an adiabatic expansion of the cloud, but, by definition, an adiabatic process does not change the phase-space density (or equivalently the average number of atoms in each energy level of the system). Thus the parameter of overriding importance in trapped systems is the phase-space density rather than the temperature.

10.4 Bose–Einstein condensation

Bosons are gregarious particles that like to be together in the same state. In contrast, fermions refuse to go into an already occupied state, e.g. electrons obey the Pauli exclusion principle (which governs the struc-

ture of atoms). Statistical mechanics tells us that when a system of bosons reaches a critical phase-space density it undergoes a phase transition and the particles avalanche into the ground state. The standard textbook treatment of this Bose–Einstein condensation (BEC) applies almost exactly to dilute vapours of alkali metals in magnetic traps, and the relative simplicity of these systems was a strong motivation for these experiments. In comparison, superfluid helium is more complex since in this *liquid* the helium atoms interact much more strongly than the atoms in a dilute vapour.¹³ Appendix F outlines a mathematical treatment of BEC that starts from the statistical mechanics of a ‘gas’ of photons in thermodynamic equilibrium, i.e. black-body radiation; this treatment shows clearly that BEC is a completely different sort of phase transition from the ‘ordinary’ condensation of a vapour into liquid caused by attractive forces between the atoms, or molecules. Quantum statistics becomes important when the occupation of quantum states approaches unity. At lower phase-space densities the particles hardly ever try to go into the same states, so they behave as classical objects; but when the states start to get crowded the particles behave differently, in a way that depends on their spin.

As shown in Appendix F, quantum effects arise when the number density $n = \mathcal{N}/V$ reaches the value¹⁴

$$n = \frac{2.6}{\lambda_{\text{dB}}^3}, \quad (10.14)$$

where λ_{dB} is the value of the thermal de Broglie wavelength defined by

$$\lambda_{\text{dB}} = \frac{h}{\sqrt{2\pi M k_{\text{B}} T}}. \quad (10.15)$$

This definition corresponds to the usual expression $\lambda_{\text{dB}} = h/Mv$ with a speed v characteristic of the gas. Simply speaking, the de Broglie wavelength gives a measure of the delocalisation of the atoms, i.e. the size of the region in which the atom would probably be found in a measurement of its position. This uncertainty in position increases as the momentum and associated kinetic energy decrease. Quantum effects become important when λ_{dB} becomes equal to the spacing between the atoms, so that the individual particles can no longer be distinguished.¹⁵

For an ideal Bose gas at the density of liquid helium (145 kg m^{-3} at atmospheric pressure) eqns 10.14 and 10.15 predict a critical temperature of 3.1 K; this is close to the so-called λ -point at 2.2 K, where helium starts to become superfluid (see Annett 2004). The equations derived for a gas give quite accurate predictions because, although helium liquefies at 4.2 K, it has a lower density than other liquids (cf. 10^3 kg m^{-3} for water). Helium atoms have weak interactions because of their atomic structure—the closed shell of electrons leads to a small size and very low polarizability. The detailed properties of superfluid helium are, however, far from those of a weakly-interacting Bose-condensed gas. In contrast, trapped atomic gases have much lower densities so BEC occurs at temperatures of around one microkelvin. It is quite amazing that sodium

¹³Historically, the idea of BEC arose after the Indian physicist Satyendra Bose published a paper in 1924 that derived the Planck distribution for radiation in a new way, by looking at a statistical distribution of photons over the energy levels. Einstein realised that the same approach could be applied to particles (that we now call bosons), and he predicted the occurrence of Bose–Einstein condensation. Einstein wrote, ‘the theory is pretty but there may be some truth to it’, in a letter to Paul Ehrenfest.

¹⁴The symbol n is used for number density (as in most statistical mechanics texts) rather than N as in previous chapters. The number of atoms is $\mathcal{N}$ and V is the volume.

¹⁵This very general criterion also applies to fermions but not with the same numerical coefficient as in eqn 10.14.

¹⁶In these systems the Bose-condensed gas is metastable; however, conditions can be reached where the condensate has a lifetime of many minutes. The ultra-cold molecules that form when cold atoms recombine are interesting to study in their own right. To achieve long lifetimes, limited by the recombination of atoms, the condensate must be held in an ultra-high vacuum to reduce the rate of collisions with molecules of the background gas in the apparatus.

and other alkalis can exist as atomic vapours at such low temperatures, and indeed this was doubted by many people before it was achieved in experiments. This is possible at very low densities because the processes that lead to recombination into molecules, and also heat the sample, occur slowly compared to the formation of a Bose–Einstein condensate.¹⁶

10.5 Bose–Einstein condensation in trapped atomic vapours

The usual textbook discussions of Bose–Einstein condensation consider a gas with a uniform density throughout a box of fixed volume, i.e. a homogeneous gas. However, the experiments with magnetically-trapped atoms correspond to a Bose gas in a harmonic potential and here we look at this inhomogeneous system. This section shows how to derive rough values of the important quantities in a way that gives a good physical understanding of the properties of the trapped Bose gas. A cloud of thermal atoms (i.e. not Bose-condensed) in a harmonic potential with a mean oscillation frequency ω has a radius r given by

$$\frac{1}{2}M\omega^2r^2 \simeq \frac{1}{2}k_B T. \quad (10.16)$$

To the level of accuracy required we take the volume of the cloud as $V \simeq 4r^3$ (a reasonable approximation to the volume of a sphere $4\pi r^3/3$). This gives the number density as $n \simeq \mathcal{N}/4r^3$, which, when combined with eqn 10.14, gives

$$\mathcal{N}^{1/3} \simeq \frac{r}{\lambda_{dB}} = \frac{k_B T_C}{\hbar\omega}. \quad (10.17)$$

¹⁷Neglecting a factor of $10^{1/3} \simeq 2$.

¹⁸This number at T_C has been chosen because after further evaporation it would lead to a condensate of roughly 10^6 atoms.

¹⁹Once a few bosons have accumulated in a particular state others want to join them. This cooperative behaviour arises because the constructive interference for bosons leads to a rate of stimulated transitions into a level proportional to the number in that level.

²⁰This number of atoms gives a T_C close to the recoil limit for sodium (and T_C varies only slowly with $\mathcal{N}$). Hence the atoms have a de Broglie wavelength comparable with that of laser cooling light. This *coincidence* gives a convenient way of remembering the approximate values. Note that T_C depends on the species of atom, since in a trap with a given magnetic field, limited by the maximum current through the coils, the oscillation frequencies of atoms are inversely proportional to $\sqrt{M}$.

This result comes from substituting for r from eqn 10.16 and for λ_{dB} at the critical temperature, T_C , from eqn 10.15.¹⁷ Fortunately, despite all our approximations, this expression lies within 10% of the value derived more carefully for a Gaussian distribution of atoms (see books by Pethick and Smith (2001) and Pitaevskii and Stringari (2003)). When the trapping potential does not have spherical symmetry, this result can be adapted by using the geometrical mean

$$\bar{\omega} = (\omega_x \omega_y \omega_z)^{1/3}. \quad (10.18)$$

For a cloud of $\mathcal{N} = 4 \times 10^6$ atoms we find¹⁸

$$k_B T_C = \hbar \bar{\omega} \mathcal{N}^{1/3} = \hbar \bar{\omega} \times 160. \quad (10.19)$$

This result shows clearly that at the BEC transition the atoms occupy many levels of the trap and that it is quantum statistics which causes atoms to avalanche into the ground state.¹⁹ A typical trap with $\bar{\omega}/2\pi = 100$ Hz has a level spacing of $\hbar \bar{\omega}/k_B \equiv 5$ nK (in temperature units). In this trap $T_C \simeq 1 \mu\text{K}$.²⁰ For 4×10^6 sodium atoms in this trap, eqn 10.14 gives as the density at T_C

$$n_C \simeq 40 \mu\text{m}^{-3} \equiv 4 \times 10^{13} \text{ cm}^{-3}.$$

The quantum statistics of identical particles applies to composite particles in the same way as for elementary particles, so long as the internal degrees of freedom are not excited. This condition is well satisfied for cold atoms since the energy required to excite the atomic electrons is much greater than the interaction energy.²¹

10.5.1 The scattering length

The quantum theory of scattering is well described in most quantum mechanics texts, and the brief summary given here is only intended as a reminder of the salient points that are relevant for understanding the collisions between ultra-cold atoms in a gas. An important feature of very low-energy collisions is that, although the potential of the attractive interaction between two atoms has the shape shown in Fig. 10.7, the overall effect is the same as a hard-sphere potential. Thus we can model a low-temperature cloud of atoms as a gas of hard spheres,²² in particular for the calculation of the contribution to the energy of the gas from interactions between the atoms.

This section gives a justification for this behaviour of ultra-cold atoms by simple physical arguments without mathematical details. The molecular potential (as in Fig. 10.7) has bound states that correspond to a diatomic molecule formed by the two atoms, and this part of the molecular wavefunction is a standing wave analogous to those that lead to the quantised energy levels of electrons in atoms.²³ It is the unbound states, however, that are appropriate for describing collisions between atoms in a gas and these correspond to travelling-wave solutions of the Schrödinger equation (illustrated in Figs 10.8 and 10.9). In the quantum mechanical treatment we need to solve the Schrödinger equation to determine what happens to an atomic wavepacket in the potential. The angular momentum of a particle in a radial potential $V(r)$ is a conserved

²¹P. Ehrenfest and J. R. Oppenheimer proved the spin-statistics relation for indistinguishable composite particles in ‘Note on the statistics of nuclei’, *Phys. Rev.*, **37**, 333 (1931).

²²At least in most of the cases of interest for ultra-cold atoms.

²³This part of the molecular wavefunction represents the vibrational motion of the molecule. Other aspects of molecular physics are described in Atkins (1994).

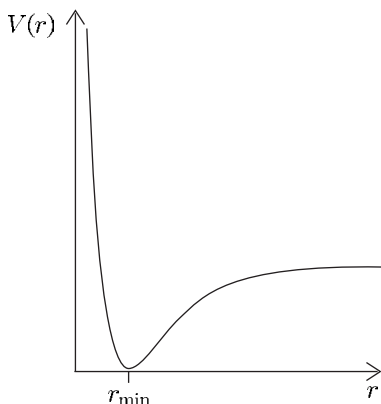


Fig. 10.7 The potential $V(r)$ for the interaction between two neutral atoms as a function of their separation r . At small separations the shells of electrons around each atom overlap and the strong electrostatic repulsion keeps the atoms apart, while at larger separations attractive van der Waals interactions dominate. These forces balance each other at the separation, where the potential is a minimum—this corresponds to the equilibrium separation of the diatomic molecule that is the bound state of the two atoms, e.g. the two sodium atoms are 0.3 nm apart in Na_2 . This interatomic potential is called the molecular potential and it determines other properties of the molecule, e.g. its vibrational frequency (corresponding to oscillations in the potential well).

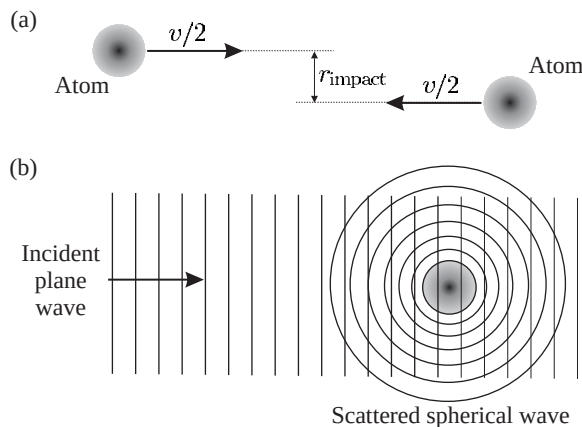
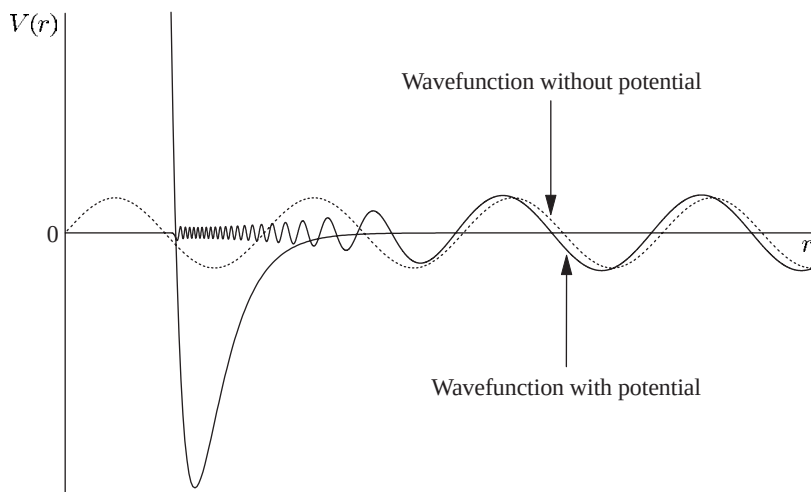


Fig. 10.8 (a) A pair of colliding atoms with relative velocity v in their centre-of-mass frame. The impact parameter r_{impact} determines their relative orbital angular momentum (which is conserved). (b) In the quantum mechanical description of a low-energy scattering the solution of the Schrödinger equation is the sum of an incident plane wave e^{ikz} plus the wave scattered by the potential that expands outwards from $r = 0$, i.e. a wavefunction of the form $\psi \propto e^{ikz} + f_k(\theta) e^{ikr}/r$. This is an eigenstate with energy $E = \hbar^2 k^2/M$ (for elastic scattering, i.e. no loss of energy). Only the scattering amplitude $f_k(\theta)$ depends on the potential $V(r)$. For low energies the scattering amplitude is a constant, $f \propto Y_{0,0}$, and the condition in eqn 10.20 is fulfilled so that the phases kz and kr have a negligible variation across the region of interaction; hence $\psi \simeq 1 + f/r$. Writing the scattering amplitude as $f = -a$ so that $\psi(r) = 0$ on a spherical surface of radius $r = a$, shows that the scattered wave is a spherical wave that is equivalent to the scattering from a hard sphere of this radius. The comparison with a hard sphere is useful for positive values of a , but scattering theory allows negative values $a < 0$ for which the outgoing wave is also spherical.

Fig. 10.9 The solution of the Schrödinger equation for low-energy scattering from a molecular potential (cf. Fig. 10.7). The plot shows $P(r)$, where $R(r) = P(r)/r$ is the radial wavefunction. At long range the overall effect of any potential (with a finite range) on the scattered wave is a phase shift (relative to a wave scattered from a point-like object at $r = 0$, shown as a dotted line)—in this region $P(r) = \sin(kr - \phi)$ and the scattering is indistinguishable from that of a hard-sphere potential that gives the same phase shift. Potentials are characterised by the radius a of the equivalent hard-sphere potential. Radial wavefunctions can be calculated by the numerical method outlined in Exercise 4.10, and the extension of such a computational approach to study quantum scattering is described in Greenhow (1990). Figure from Butcher *et al.* (1999).



quantity, in both classical and quantum mechanics; thus the eigenstates of the angular part of the Schrödinger equation are the spherical harmonic functions, as in the central-field approximation for atoms. We can deduce the orbital angular momentum by correspondence with the following classical calculation.

A pair of colliding atoms has relative orbital angular momentum $\hbar l \simeq M' v r_{\text{impact}}$, where M' is the reduced mass,²⁴ v is their relative velocity and r_{impact} is the impact parameter (defined in Fig. 10.8). For a collision to happen r_{impact} must be less than the range of the interaction r_{int} . Thus we find that $\hbar l \lesssim M' v r_{\text{int}} = \hbar r_{\text{int}} / \lambda_{\text{dB}}$, using the de Broglie relation. This implies that $l \lesssim 2\pi r_{\text{int}} / \lambda_{\text{dB}}$ and therefore, when the energy is sufficiently low that

$$\frac{\lambda_{\text{dB}}}{2\pi} \gg r_{\text{int}}, \quad (10.20)$$

we have $l = 0$, i.e. the atoms have no relative orbital angular momentum. In this regime, the scattered wavefunction is a spherical wave proportional to $Y_{l=0,m=0}$ no matter how complicated the actual potential. At the recoil limit of laser cooling the atoms and the photons of the laser light have a comparable wavelength $\lambda_{\text{dB}} \simeq \lambda_{\text{light}}$, because they have similar momentum, e.g. sodium atoms at $T_{\text{recoil}} = 2 \mu\text{K}$ have $\lambda_{\text{dB}}/2\pi \simeq 100 \text{ nm}$.²⁵ This estimate indicates that the condition 10.20 is fulfilled at temperatures of a few microkelvin since the range of the interaction between neutral atoms is normally considerably less than 100 nm (equivalent to 2000 Bohr radii).²⁶ This spherical wave corresponding to the eigenfunction $Y_{0,0}$ is called the s-wave, where s denotes zero relative orbital angular momentum (cf. s-orbitals that are bound states with $l = 0$).²⁷

The discussion of the s-wave scattering regime justifies the first part of the statement above that low-energy scattering from any potential looks the same as scattering from a hard-sphere potential when the radius of the sphere is chosen to give the same strength of scattering. The radius of this hard sphere is equivalent to a parameter that is usually called the **scattering length** a . This single parameter characterises the low-energy scattering from a particular potential.²⁸ For example, sodium atoms in the $|F = 1, M_F = 1\rangle$ state have $a = 2.9 \text{ nm}$, which is about an order of magnitude greater than the size of the atom's electronic charge cloud and does not correspond to any physical feature in the real atom. In the following, the energy contribution from interactions between the atoms in a low-temperature gas is calculated assuming that the atoms act like hard spheres (which is just a useful fiction that is mathematically equivalent to the scattering from the actual potential). First, let us study a simple example that illustrates features that arise in the general case.

Example 10.3 A particle in a spherical well

The Schrödinger equation for a particle in a spherically-symmetric potential can be separated into an angular equation and a radial equation that can be written in terms of $P(r) = rR(r)$, as in eqn 2.16. In this

²⁴The difference between the reduced mass and the mass of the individual atoms is not particularly important in this rough estimate.

²⁵Calculated in the laboratory frame of reference, i.e. not using the reduced mass.

²⁶Molecular potentials, such as that shown in Fig. 10.7, do not have a sharply-defined cut-off. The determination of the minimum distance at which the atoms can pass with a negligible effect on each other requires a more general treatment.

²⁷In the particular case of two identical bosons in the same internal state, the spatial wavefunction must be symmetric with respect to an interchange of the particle labels. Such wavefunctions have even orbital angular momentum quantum numbers $l = 0, 2, 4$, etc. Thus p-wave scattering cannot occur for collisions between identical bosons and the s-wave scattering regime extends up to the threshold energy for d-waves.

²⁸There are many potentials that can give the same value of a .

example we shall examine the properties of wavefunctions with $l = 0$ (corresponding to s-waves), so that the equation for $P(r)$ is simply

$$\left[-\frac{\hbar^2}{2M'} \frac{d^2}{dr^2} + V(r) \right] P(r) = EP(r), \quad (10.21)$$

where M' is the mass of the particle. For the potential $V(r) = 0$ within the range $a \leq r \leq b$ and $V(r) = \infty$ elsewhere (for $r < a$ and $r > b$), this equation has the same form as that for an infinite square well in one dimension. The solution that satisfies the boundary condition $\psi(r) = 0$ at $r = a$ is

$$P = C \sin(k(r - a)), \quad (10.22)$$

where C is an arbitrary constant. The boundary condition that the wavefunction is zero at $r = b$ requires that $k(b - a) = n\pi$, where n is an integer; hence the energy eigenvalues are given by²⁹

$$E = \frac{\hbar^2 k^2}{2M'} = \frac{\hbar^2 \pi^2 n^2}{2M'(b - a)^2}. \quad (10.23)$$

To make a link with scattering theory we consider what happens when $a \ll b$, so that the wavefunction is contained in a spherical region of radius b but excluded from a small hard sphere of radius a at the origin; the energy of the lowest level ($n = 1$) can be written as

$$E = \frac{\hbar^2 \pi^2}{2M'b^2} \left(1 - \frac{a}{b}\right)^{-2} \simeq E(a = 0) + \frac{\hbar^2 \pi^2 a}{M'b^3}. \quad (10.24)$$

This equals the energy for $a = 0$ plus a small perturbation proportional to a which arises because the kinetic energy depends on the size of the region between $r = a$ and b . (The expectation value of the potential energy is zero.) At short range where $\sin(k(r - a)) \simeq k(r - a)$ the solution in eqn 10.22 reduces to

$$R(r) \simeq \frac{P(r)}{r} \propto 1 - \frac{a}{r}. \quad (10.25)$$

This is the general form for a wavefunction with a low energy ($ka \ll 1$) near a hard sphere (in the region $a < r \ll \lambda_{dB}/2\pi$) and the features illustrated by this example arise in the general case. The use of this situation to illustrate scattering might seem to contradict the above assertion that scattering involves wavefunctions that are unbound states, but we will see below that such waves have a similar increase in energy proportional to a . In any case, we will be applying the results to atoms that are confined (in the harmonic potential created by a magnetic trap).

A collision between a pair of atoms is described in their centre-of-mass frame as the scattering from a potential $V(r)$ of a particle with a reduced mass given by³⁰

$$M' = \frac{M_1 M_2}{M_1 + M_2}. \quad (10.26)$$

²⁹This is the usual result for an infinite square well of length L rewritten with $L = b - a$.

³⁰The reduced mass is used for the same reason as in the hydrogen atom, i.e. in one-electron atoms the electron and nucleus orbit around their centre-of-mass frame, and therefore it is the reduced mass of the electron $m_e \times M_N/(m_e + M_N)$ that appears in the Schrödinger equation for the atom—the (slight) dependence of the electron's reduced mass on the nuclear mass M_N leads to the isotope shift of spectral lines (Chapter 6).

In a gas of identical particles, the two colliding atoms have the same mass $M_1 = M_2 = M$ and therefore their reduced mass is $M' = M/2$.³¹

The discussion above showed that, in the s-wave regime, the scattering is the same for a hard-sphere potential and for the actual molecular potential (see also Fig. 10.9). Using the wavefunction in eqn 10.25, with an amplitude χ , we find that the expectation value of the kinetic energy operator $(-\hbar^2/2M')\nabla^2$, with $M' = M/2$, is given by³²

$$\begin{aligned} E_a &= \iiint \frac{\hbar^2}{M} \left| \nabla \left\{ \chi \left(1 - \frac{a}{r} \right) \right\} \right|^2 d^3\mathbf{r} \\ &= -\frac{4\pi\hbar^2}{M} |\chi|^2 \int_a^\infty \left| \frac{d}{dr} \left(1 - \frac{a}{r} \right) \right|^2 r^2 dr \\ &= \frac{4\pi\hbar^2 a}{M} |\chi|^2. \end{aligned} \quad (10.27)$$

Taking the upper limit of integration over r as infinity gives a reasonable estimate of the energy (Pathra 1971).³³ This increase in energy caused by the interaction between atoms has the same scaling with a as in eqn 10.24, and arises from the same physical origin. We shall use this result to account for the interatomic interactions in a Bose–Einstein condensate in the following section.

Finally, we note a subtle modification of scattering theory for identical particles. The usual formula for the collision cross-section is $4\pi a^2$, but identical bosons have

$$\sigma = 8\pi a^2. \quad (10.28)$$

The additional factor of 2 arises because bosons constructively interfere with each other in a way that enhances the scattering.³⁴ This and the other features of collisions between ultra-cold bosons relevant to Bose–Einstein condensation are explained more fully in the books by Metcalf and van der Straten (1999), Pethick and Smith (2001), and Pitaevskii and Stringari (2003). These references also give a more careful definition of the scattering length that shows why this parameter can be positive $a > 0$, or negative $a < 0$ (see Fig. 10.10). However, the majority of experimental work on Bose–Einstein condensation has been carried out with states of sodium and rubidium atoms that have positive scattering lengths, corresponding to the effectively repulsive hard-sphere interactions considered in this section. A full treatment of quantum scattering also allows a more rigorous derivation of the temperature below which there is only s-wave scattering, but the exact limits of the s-wave regime are not important for the purposes of this chapter since collisions between atoms at temperatures of a few microkelvin are normally well within this regime.³⁵

³¹The transformation to the centre-of-mass frame and the use of reduced mass is very similar in classical and quantum mechanics.

³²The integrand $|\nabla\psi|^2 = \nabla\psi^*\nabla\psi$ is obtained from $\psi^*\nabla^2\psi$ by integration by parts (as in the standard derivation of probability current in quantum mechanics).

³³Most of the contribution to the integral comes from the region where $a < r \ll \lambda_{dB}/2\pi$, where eqn 10.25 is a good approximation to the wavefunction.

³⁴The probability of bosons going into a particular quantum state is enhanced by a factor $\mathcal{N} + 1$, where $\mathcal{N}$ is the number of particles in that state—for a two-body collision this increases the probability by a factor of 2. In contrast, two identical fermions cannot occupy the same state and therefore s-wave collisions, in which the particles have the same spatial state, are forbidden for fermions in the same spin state. (As noted previously, p-wave collisions do not occur for bosons.)

³⁵Exceptions might occur where a resonance gives especially strong interactions between the atoms.

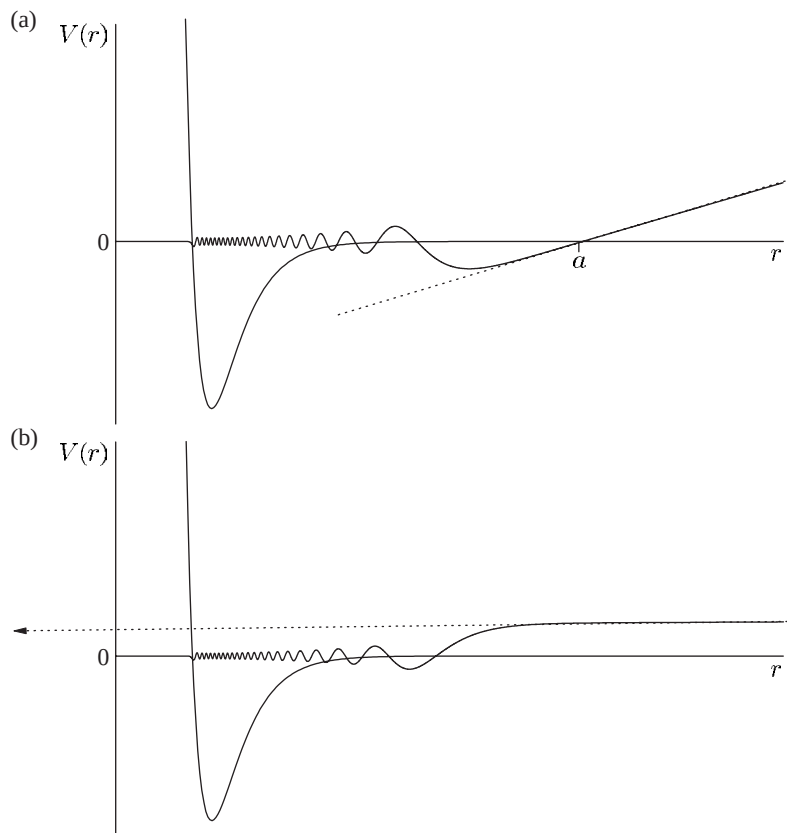


Fig. 10.10 The wavefunction $P(r) = rR(r)$ for very-low-energy scattering from slightly different molecular potentials with scattering lengths that are (a) positive and (b) negative with very large magnitude. From Butcher *et al.* (1999). The extrapolation of the wavefunction from at large r is drawn as a dashed line that crosses the horizontal axis at $r = a$.

10.6 A Bose–Einstein condensate

The interaction between atoms is taken into account by including a term in the Schrödinger equation that comes from eqn 10.27, proportional to the square of the wavefunction:

$$\left\{ -\frac{\hbar^2}{2M} \nabla^2 + V(\mathbf{r}) + g|\psi|^2 \right\} \psi = \mu\psi. \quad (10.29)$$

The extra energy from the interactions is proportional to $|\psi|^2$, the probability of finding a particle in a given region, and the coupling constant is

$$g = \frac{4\pi\hbar^2\mathcal{N}a}{M}. \quad (10.30)$$

This comes from eqn 10.27 with $|\chi|^2 \rightarrow \mathcal{N}|\psi|^2$ that gives the interaction per atom in the presence of $\mathcal{N}$ atoms.³⁶ The symbol μ is used (instead of E) to represent the energy of an individual atom in the presence of all the others (cf. the central-field approximation in Chapter 4).³⁷ This nonlinear Schrödinger equation is called the Gross–Pitaevskii equation after the people who first (independently) wrote it down and a rigorous

³⁶ Actually, there are $\mathcal{N} - 1$ other atoms, but the difference from $\mathcal{N}$ is negligible for large numbers of atoms.

³⁷ This quantity turns out to be equivalent to the chemical potential in thermodynamics: $\mu = \partial E / \partial \mathcal{N}$ is the energy required to remove a particle from the system. This is not the same as the average energy per particle—see Exercise 10.7.

derivation can be found in the book by Pitaevskii and Stringari (2003). Trapped atoms experience a harmonic potential

$$V(\mathbf{r}) = \frac{1}{2}M(\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2). \quad (10.31)$$

For simplicity, we shall consider all three oscillation frequencies equal, i.e. the isotropic potential $M\omega^2 r^2/2$, and use a variational method to estimate the energy. We choose a trial wavefunction that is a Gaussian function:

$$\psi = Ae^{-r^2/2b^2}. \quad (10.32)$$

Using this to calculate the expectation values of terms in eqn 10.29 gives

$$E = \frac{3}{4}\hbar\omega \left\{ \frac{a_{\text{ho}}^2}{b^2} + \frac{b^2}{a_{\text{ho}}^2} \right\} + \frac{g}{(2\pi)^{3/2}} \frac{1}{b^3}. \quad (10.33)$$

Differentiation of this expression shows that when $g = 0$ the minimum energy occurs when $b = a_{\text{ho}}$, where

$$a_{\text{ho}} = \sqrt{\frac{\hbar}{M\omega}} \quad (10.34)$$

is the characteristic radius of the Gaussian ground-state wavefunction in the quantum harmonic oscillator. A sodium atom in a trap with oscillation frequency $\omega/2\pi = 100$ Hz has $a_{\text{ho}} = 2 \times 10^{-6}$ m. For this equilibrium value of b , the two terms representing the kinetic and potential energies give an equal contribution to the total energy which is $E = (3/2)\hbar\omega$, as expected for the ground state of the quantum harmonic oscillator.³⁸ The variational method gives exact results in this particular case because the trial wavefunction has the same Gaussian form as the actual solution for a harmonic oscillator.

Now we shall consider what happens when $g > 0$.³⁹ The ratio of the terms representing the atomic interactions and the kinetic energy is⁴⁰

$$\frac{4}{3(2\pi)^{3/2}} \frac{g}{a_{\text{ho}}^3 \hbar\omega} \simeq \frac{\mathcal{N}a}{a_{\text{ho}}}. \quad (10.35)$$

The nonlinear term swamps the kinetic energy when $\mathcal{N} > a_{\text{ho}}/a$. This ratio equals $a_{\text{ho}}/a = 700$ for $a_{\text{ho}} = 2 \mu\text{m}$ and $a = 3$ nm, and so in this case when the number of atoms in the condensate $\mathcal{N} > 700$ we can neglect the kinetic energy.⁴¹ We could estimate the condensate's size from eqn 10.33 by determining the value of r at which the confining potential balances the repulsive interactions, and this variational method is described in Exercise 10.6; but it is remarkably easy to solve the Gross–Pitaevskii equation when the kinetic energy term is neglected. In this so-called Thomas–Fermi regime eqn 10.29 becomes simply

$$\{V(r) + g|\psi|^2\}\psi = \mu\psi. \quad (10.36)$$

So for the region where $\psi \neq 0$ we find

$$|\psi|^2 = \frac{\mu - V(r)}{g}. \quad (10.37)$$

³⁸Each of the three degrees of freedom has a zero-point energy of $\frac{1}{2}\hbar\omega$.

³⁹Small Bose–Einstein condensates with effectively attractive interactions have been created, but they collapse inward as the number of atoms grows (see Exercise 10.11).

⁴⁰The factor of $(4/3)(4\pi/(2\pi)^{3/2})$ that arises has a numerical value of about unity.

⁴¹Typically, experiments have $\mathcal{N} > 10^5$ and the effectively repulsive interactions make the condensate much larger than a_{ho} . The interactions between the atoms in a dilute gas only have a small effect ($< 10\%$) on the value of T_{C} .

Hence, the number density of atoms $n(r) = \mathcal{N}|\psi(r)|^2$ in the harmonic potential has the form of an inverted parabola:

$$n(r) = n_0 \left(1 - \frac{x^2}{R_x^2} - \frac{y^2}{R_y^2} - \frac{z^2}{R_z^2} \right), \quad (10.38)$$

where n_0 , the number density at the centre of the condensate, is

$$n_0 = \frac{\mathcal{N}\mu}{g}. \quad (10.39)$$

The condensate has an ellipsoidal shape and the density goes to zero at points on the axes given by $x = \pm R_x$, $y = \pm R_y$ and $z = \pm R_z$, defined by

$$\frac{1}{2}M\omega_x^2 R_x^2 = \mu, \quad (10.40)$$

and similarly for R_y and R_z . In the Thomas–Fermi regime, the atoms fill up the trap to the level of the chemical potential as illustrated in Fig. 10.11, just like water in a trough. The chemical potential μ is determined by the normalisation condition

$$1 = \iiint |\psi|^2 dx dy dz = \frac{\mu}{g} \frac{8\pi}{15} R_x R_y R_z. \quad (10.41)$$

A useful form for μ is

$$\mu = \hbar\bar{\omega} \times \frac{1}{2} \left(\frac{15\mathcal{N}a}{a_{\text{ho}}} \right)^{2/5}. \quad (10.42)$$

The mean oscillation frequency is defined as $\bar{\omega} = (\omega_x\omega_y\omega_z)^{1/3}$ and a_{ho} is calculated using this frequency in eqn 10.34. Typical values for an Ioffe trap are given in the following table together with the important properties of a BEC of sodium, calculated from the key formulae in eqns 10.42, 10.40 and 10.39 (in that order).

Scattering length (for Na)	a	2.9 nm
Radial oscillation frequency ($\omega_x = \omega_y$)	$\omega_x/2\pi$	250 Hz
Axial oscillation frequency	$\omega_z/2\pi$	16 Hz
Average oscillation frequency	$\bar{\omega}/2\pi$	100 Hz
Zero-point energy (in temperature units)	$\frac{1}{2}\hbar\bar{\omega}/k_B$	2.4 nK
Harmonic oscillator length (for $\bar{\omega}$)	a_{ho}	2.1 μm
Number of atoms in condensate	$\mathcal{N}_0$	10^6
Chemical potential	μ	130 nK
Radial size of condensate	$R_x = R_y$	15 μm
Axial size of condensate	R_z	95 μm
Peak density of condensate	n_0	$2 \times 10^{14} \text{ cm}^{-3}$
Critical temperature (for 4×10^6 atoms)	T_C	760 nK
Critical density (for 4×10^6 atoms)	n_C	$4 \times 10^{13} \text{ cm}^{-3}$

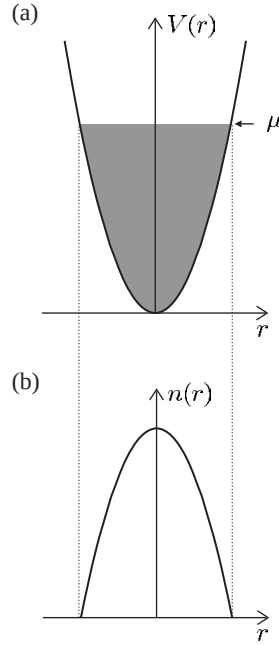


Fig. 10.11 In the Thomas–Fermi regime the condensate has the same shape as the confining potential. (a) A harmonic potential. (b) The density of the atoms in a harmonic trap has an inverted-parabolic shape (along all three axes).

The critical temperature and density at the onset of Bose condensation are not properties of the condensate, but are calculated for a cloud of 4×10^6 atoms (from eqn 10.19); this would lead to a condensate of roughly $\mathcal{N}_0 \simeq 10^6$ atoms after evaporative cooling to $T/T_C \simeq 0.5$ (where most atoms are in the condensate, see eqn F.16). Both μ and T_C have a weak dependence on $\mathcal{N}$ (eqns 10.19 and 10.42) and a similar (but not the same) dependence on ω , so these quantities have a similar relative magnitude for many cases. Note, however, that μ depends on

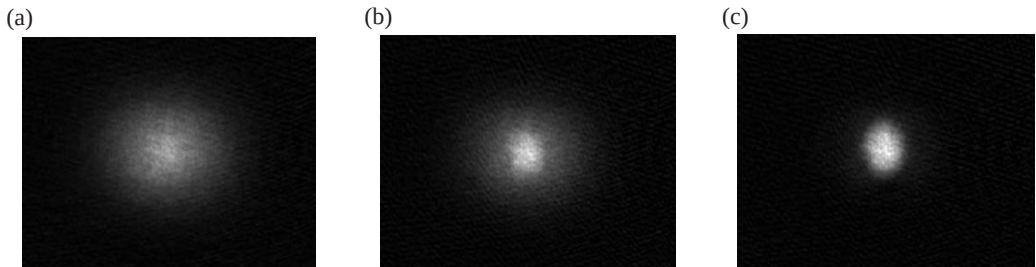


Fig. 10.12 This sequence of images shows a Bose–Einstein condensate being born out of a cloud of evaporatively-cooled atoms in a magnetic trap. Each image was taken after a time-of-flight expansion. The cloud changes its size and shape as it undergoes a phase transition: (a) a thermal cloud just above the critical temperature T_C has a spherical shape (isotropic expansion); (b) a cloud of atoms at $0.9 T_C$ has a Bose-condensed fraction in the centre surrounded by a halo of thermal atoms; and (c) well below the critical temperature ($< 0.5 T_C$) most of the atoms are in the condensate (lowest energy state of the trap). These images come from a system that does not have the same aspect ratio as an Ioffe trap, but they illustrate the anisotropic expansion of the condensate wavefunction. See Fig. 10.13 for dimensions and other details. Data provided by Nathan Smith, Physics department, University of Oxford.

⁴²The repulsive interactions prevent the much greater increase in the density that would occur if atoms congregated in a region of volume a_{ho}^3 .

the strength of the interactions, whereas T_C does not. In this example the condensate has a density about a factor of 5 greater than the thermal cloud at the phase transition,⁴² but the gas remains dilute because the average distance between atoms in the condensate is larger than the scattering length, that is $na^3 \ll 1$ (for the data in the table $n_0 a^3 = 4 \times 10^{-6}$). Equation 10.40, and the similar equation for R_z , give the ratio of sizes as $R_z/R_x = \omega_x/\omega_z = 16$ (and $R_y = R_x$), so in this trap the condensate has the shape of a long, thin cigar.

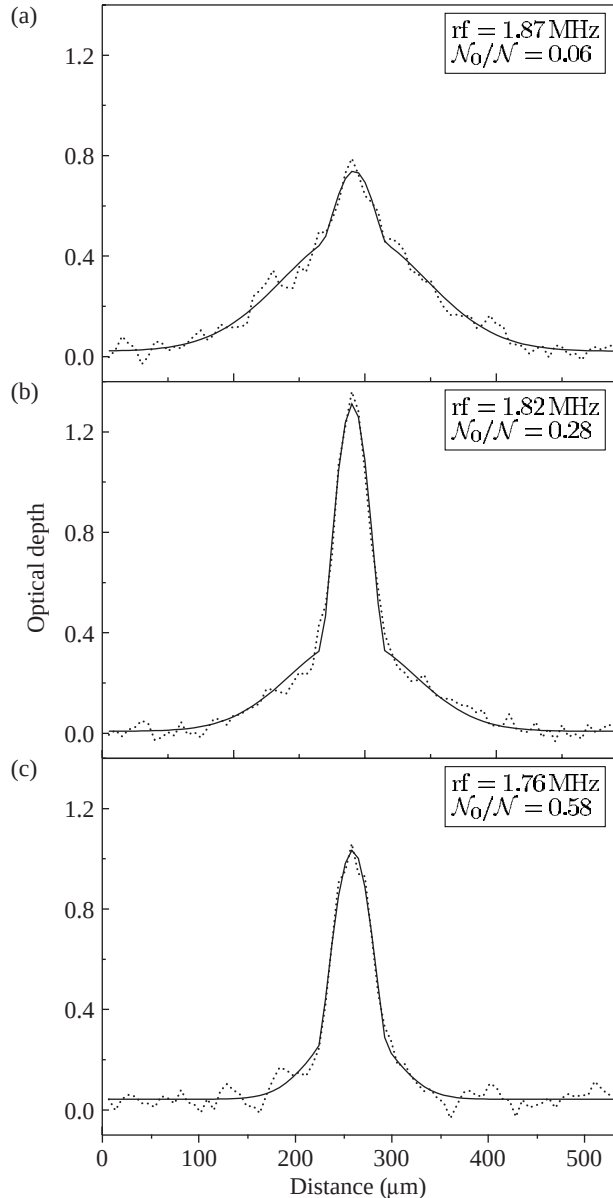


Fig. 10.13 Cross-sections of the images similar to those shown in Fig. 10.12, but for different temperatures and a time of 12 ms after release from the magnetic trap. (a) Just below the critical temperature ($0.99 T_C$) a small central peak appears on the Gaussian distribution of thermal atoms. (b) At $0.82 T_C$ most of the atoms are in the condensate with some thermal atoms in the wings. (c) At $0.63 T_C$ only a small thermal cloud remains. This narrowing of the distribution and change from a Gaussian to an inverted-parabolic shape occurs over a small range of temperatures which is a behaviour characteristic of a phase transition. From Hechenblaikner (2002), for a trap with $\omega_x = \omega_y = 2\pi \times 126$ Hz and $\omega_z = 2\pi \times 356$ Hz. The fraction of atoms in the condensate (N_0/N) differs from that predicted by eqn F.16 because of interactions (see Maragò *et al.* 2001).

To observe the condensate experimenters record an image by illuminating the atoms with laser light at the resonance frequency.⁴³ Typically, the experiments have an optical resolution of about $5\text{ }\mu\text{m}$, so that the length of the condensate can be measured directly but its width is not precisely determined. Therefore the magnetic trap is turned off sharply so that the atoms expand and some time later a laser beam, that passes through the cloud of atoms onto a camera, is flashed on to record a shadow image of the cloud. The repulsion between atoms causes the cloud to expand rapidly after the confining potential is switched off (see Exercise 10.6). The cigar-shaped cloud expands more rapidly in the radial direction (x and y) than along z , so that after several milliseconds the radial size becomes bigger than that along z , i.e. the aspect ratio inverts.⁴⁴ In contrast, the uncondensed atoms behave as a classical gas and expand isotropically to give a spherical cloud, since by definition the thermal equilibrium implies the same kinetic energy in each direction. Pictures such as Fig. 10.12 are the projection of the density distribution onto a two-dimensional plane, and show an obvious difference in shape between the elliptical condensate and the circular image of the thermal atoms. This characteristic shape was one of the key pieces of evidence for BEC in the first experiment, and it is still commonly used as a diagnostic in such experiments. Figure 10.13 shows the density profile of the cloud of atoms released from a magnetic trap for temperatures close to the critical point, and below.

⁴³Generally, absorption gives a better signal than fluorescence but the optical system and camera are similar in both cases.

⁴⁴This expansion of the wavefunction is predicted by including time dependence in the nonlinear Schrödinger equation.

10.7 Properties of Bose-condensed gases

Two striking features of Bose-condensed systems are superfluidity and coherence. Both relate to the microscopic description of the condensate as $\mathcal{N}$ atoms sharing the same wavefunction, and for Bose-condensed gases they can be described relatively simply from first principles (as in this section). In contrast, the phenomena that occur in superfluid helium are more complex and the theory of quantum fluids is outside the scope of this book.

10.7.1 Speed of sound

To estimate the speed of sound v_s by a simple dimensional argument we assume that it depends on the three parameters μ , M and ω , so that⁴⁵

$$v_s \propto \mu^\alpha M^\beta \omega^\gamma. \quad (10.43)$$

This dimensional analysis gives⁴⁶

$$v_s \simeq \sqrt{\frac{\mu}{M}}. \quad (10.44)$$

This corresponds to the actual result for a homogeneous gas (without us needing to insert any numerical factor), and gives a fairly good approximation in a trapped sample. The speed at which compression waves

⁴⁵The size of the condensate R is not another independent parameter, see eqn 10.40.

⁴⁶Comparing the dimensions of the terms in eqn 10.43 gives

$$\text{m s}^{-1} = [\text{kg m}^2 \text{s}^{-2}]^\alpha \text{kg}^\beta \text{s}^{-2\gamma}.$$

Hence $\alpha = -\beta = 1/2$ and $\gamma = 0$.

travel in the gas has great significance for superfluidity. For motion slower than this speed the condensate flows smoothly around obstacles without exciting any particles out of the ground state of the quantum gases. This type of flow does not dissipate any energy and so it is frictionless and the gas is superfluid.

10.7.2 Healing length

The Thomas–Fermi approximation neglects the kinetic energy term in the Schrödinger equation. This leads to a physically unrealistic sharp edge at the surface of the condensate (see Fig. 10.11)—such a discontinuity in the gradient would make $\nabla^2\psi$ infinite. Therefore we have to take kinetic energy into account at the boundary. To determine the shortest distance ξ over which the wavefunction can change we equate the kinetic term (that contains $\nabla^2\psi \simeq \hbar^2/(2M\xi^2)$) to the energy scale of the system given by the chemical potential. Atoms with energy higher than μ leave the condensate. Using $n_0 = \mathcal{N}\mu/g$ (from eqn 10.39) and eqn 10.30 for g , we find that

$$\frac{\hbar^2}{2M\xi^2} \simeq \mu = \frac{gn_0}{\mathcal{N}} = \frac{4\pi\hbar^2an_0}{M}. \quad (10.45)$$

Hence $\xi = 1/\sqrt{8\pi an_0}$, e.g. $\xi = 0.3\mu\text{m}$ for a sodium condensate with $n_0 = 2 \times 10^{14}\text{cm}^{-3}$. Typically, $\xi \ll R_x$ and smoothing of the wavefunction only occurs in a thin boundary layer, and these surface effects give only small corrections to results calculated using the Thomas–Fermi approximation. This so-called **healing length** also determines the size of the vortices that form in a superfluid when the confining potential rotates (or a fast moving object passes through it). In these little ‘whirlpools’ the wavefunction goes to zero at the centre, and ξ determines the distance over which the density rises back up to the value in the bulk of the condensate, i.e. this healing length is the distance over which the superfluid recovers from a sharp change.

10.7.3 The coherence of a Bose–Einstein condensate

Figure 10.14 shows the result of a remarkable experiment carried out by the group led by Wolfgang Ketterle at MIT. They created two separate condensates of sodium at the same time. After the trapping potential was turned off the repulsion between the atoms caused the two clouds to expand and overlap with each other (as in the time-of-flight technique used to observe Bose–Einstein condensation, see Fig. 10.12). The two condensates interfere to give the fringes shown in the figure; there are no atoms at certain positions where the matter waves from the two sources interfere destructively—these atoms do not disappear, but they are redistributed to positions in the fringe pattern where the matter waves add constructively. Such interference is well known in optics; however, there is a very interesting difference between this experiment

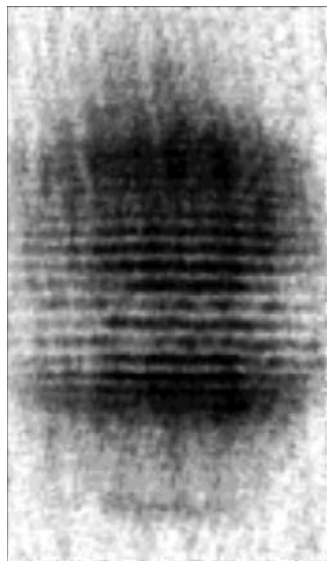


Fig. 10.14 The interference fringes observed when two independent Bose condensates are released from nearby potential wells and the clouds of atoms expand and overlap. This experiment was carried out with sodium atoms by the team led by Wolfgang Ketterle at MIT (Andrews *et al.* 1997). Copyright 1997 by the American Association for the Advancement of Science.

and the usual double-slit experiments. In the MIT experiment there was no fixed relation between the phases of the two condensates and before the experiment was carried out it was hotly debated whether interference would be observed. Clear interference fringes were observed each time the experiment was carried out. However, the position of these fringes depended on the difference between the phases of the condensates in that particular run—the bright and dark fringes appeared at a different place each time the experiment was repeated; thus the fringe pattern would ‘wash out’ if averaged over many runs. The observation of the interference of two condensates relies on the ability to see interference fringes in a single shot.

The experiment was carried out with an Ioffe trap in which the atoms form a long, cigar-shaped cloud. At MIT they used a sheet of light to chop the cloud into two pieces of roughly half the original length. The light exerted a force that pushed the atoms out of the region of high intensity (because it had a blue frequency detuning as explained in Section 9.6). This configuration gave two separate potential wells. In practice, the following situations both give the same results: (a) when the two condensates are created independently; and (b) when a single large condensate is divided into two parts after it has formed. The process of turning on a sheet of light in the middle of an already-formed condensate produces such a strong perturbation that the two resulting condensates have almost random phases. Only recently has the controlled separation of a condensate into two parts whilst preserving the phase been demonstrated in a double-well dipole-force trap—a system that corresponds to a beam splitter for matter waves. (Atom optics is discussed further in Chapter 11.) However, the intriguing aspect of the interference of two independent condensates is its dissimilarity to previous experiments.



Fig. 10.15 Atoms coupled out of a Bose–Einstein condensate fall downwards under gravity to form a well-collimated matter-wave beam, with analogous properties to the beam of light from a laser. Courtesy of Nathan Smith and William Heathcote, Physics department, University of Oxford.

10.7.4 The atom laser

The phrase ‘atom laser’ has been used to describe the coherent beam of matter waves coupled out of a Bose–Einstein condensate (as shown in Fig. 10.15). After forming the condensate, the radio-frequency radiation was tuned to a frequency that drives a transition to an untrapped state (e.g. $M_F = 0$) for atoms at a position inside the condensate. (This comes from the same source of radiation used for evaporative cooling.) These atoms fall downwards under gravity to form the beam seen in the figure. These matter waves coupled out of the condensate have a well-defined phase and wavelength like the light from a laser.⁴⁷ Many novel matter-wave experiments have been made possible by Bose–Einstein condensation, e.g. the observation of nonlinear processes analogous to nonlinear optics experiments that were made possible by the high-intensity light produced by lasers.

⁴⁷The atoms accelerate as they fall under gravity so the wave propagation is different to that of light.

10.8 Conclusions

Bose–Einstein condensation in dilute alkali vapours was first observed in 1995 by groups at JILA (in Boulder, Colorado) and at MIT, using laser cooling, magnetic trapping and evaporation. This breakthrough, and the many subsequent new experiments that it made possible, led to the award of the Nobel prize to Eric Cornell, Carl Wieman and Wolfgang Ketterle in 2001 (and the Nobel prize web site has much useful information on this subject, with links to the web sites of the research groups). Recent BEC experiments have produced a wealth of beautiful images; however, the objective of this chapter has not been to cover everything but rather to explain the general principles of the underlying physics. The two books on BEC by Pethick and Smith (2001) and Pitaevskii and Stringari (2003) contain much more detail.

Exercises

(10.1) *Magnetic trapping*

An Ioffe–Pritchard trap has a radial gradient of $b' = 3 \text{ T m}^{-1}$, and the combination of Helmholtz and pinch coils gives a field along z with $B_0 = 3 \times 10^{-4} \text{ T}$ and curvature $b'' = 300 \text{ T m}^{-2}$. Calculate the oscillation frequencies of sodium atoms in the trap.

(10.2) *Loading a trap*

- (a) A spherical cloud of 10^{10} sodium atoms with a density of around 10^{10} cm^{-3} and a temperature of $T = 2.4 \times 10^{-4} \text{ K}$ is placed in a spherically-symmetric trapping potential. The temperature and density of the cloud are preserved during this loading if

$$\frac{1}{2} M \omega^2 r^2 = \frac{1}{2} k_B T. \quad (10.46)$$

Calculate the trapping frequency ω that fulfils this mode-matching condition, and explain what happens if the trap is too stiff or too weak. (In a precise treatment r would be the root-mean-square radius of a cloud with a Gaussian density distribution.)

- (b) Calculate $n\lambda_{\text{dB}}^3/2.6$ for the trapped cloud, i.e. the ratio of its phase-space density to that required for BEC (eqn 10.14).
- (c) After loading, an adiabatic compression of the trapped cloud changes the oscillation frequencies of the atoms to $\omega_r/2\pi = 250 \text{ Hz}$ and $\omega_z/2\pi = 16 \text{ Hz}$. The phase-space density does not change during adiabatic processes, i.e. $n\lambda_{\text{dB}}^3$ is constant. Show that this implies that $TV^{2/3}$ is constant. Calculate the temperature and density of the cloud after compression.⁴⁸

(10.3) *Magnetic trapping*

- (a) Sketch the energy of the hyperfine levels of the $3s \ ^2S_{1/2}$ ground level of sodium as a function of the applied magnetic field strength. (The hyperfine-structure constant of this level is $A_{3s} = 886 \text{ MHz}$ and sodium has nuclear spin $I = 3/2$.)
- (b) What is meant by a ‘weak’ field in the context of hyperfine structure?
- (c) Show that for a weak magnetic field the states in both hyperfine levels have a splitting of 7 GHz T^{-1} .
- (d) Explain why the potential energy of an atom in a magnetic trap is proportional to the magnetic flux density $|\mathbf{B}|$. A magnetic trap has a field that can be approximated by

$$\mathbf{B} = b'(x\hat{\mathbf{e}}_x - y\hat{\mathbf{e}}_y)$$

in the region where $r = (x^2 + y^2)^{1/2} \leq 10 \text{ mm}$, and $B = 0$ outside this radius. The field gradient $b' = 1.5 \text{ T m}^{-1}$ and the z -axis of the trap is horizontal.

- (e) Calculate the ratio of the magnetic force on the atoms compared to that of gravity.
- (f) Estimate the maximum temperature of atoms that can be trapped in the (i) upper and (ii) lower hyperfine levels. State the M_F quantum number of the atoms in each case. (Assume that the confinement of atoms along the z -axis is not the limiting factor.)
- (g) For the clouds of trapped atoms in both (i) and (ii) of part (f), describe the effect of applying radio-frequency radiation with a frequency of 70 MHz .

⁴⁸The relation between temperature and volume can also be derived from thermodynamics: $TV^{\gamma-1}$ is constant for an adiabatic change in an ideal gas and a monatomic gas has a ratio of heat capacities $\gamma = C_P/C_V = 5/3$. Actually, the phase-space density only remains constant if the potential has the same shape throughout the adiabatic change. In the case of an Ioffe trap, the radial potential may change from harmonic to linear (see Example 10.2), giving a small increase in the phase-space density. This effect arises because the population of the energy levels stays the same but the distribution of the levels changes—the energy levels of a harmonic potential are equally spaced ($\hbar\omega$ apart), whereas in a linear potential the intervals between levels decrease with increasing energy.

(10.4) *Evaporative cooling*

A cloud of atoms has a Boltzmann energy distribution $\mathcal{N}(E) = Ae^{-\beta E}$, where $1/\beta = k_B T$ and the normalisation constant A is found from

$$\mathcal{N}_{\text{total}} = A \int_0^\infty e^{-\beta E} dE = \frac{A}{\beta}.$$

The cloud has a total energy given by

$$E_{\text{total}} = A \int_0^\infty E e^{-\beta E} dE = \frac{A}{\beta^2} = \mathcal{N}_{\text{total}} k_B T.$$

Hence each atom has a mean energy $\overline{E} = k_B T$. In an evaporative cooling step all atoms with energy greater than ϵ escape.

- Calculate the fraction of atoms lost $\Delta\mathcal{N}/\mathcal{N}_{\text{total}}$.
- Calculate the fractional change in the mean energy per atom.
- Evaluate your expressions for cuts with $\beta\epsilon = 3$ and 6 . Compare the ratio of energy lost and the number of atoms removed in the two cases and comment on the implications for evaporative cooling.
- The collision rate between atoms in the cloud is $R_{\text{coll}} = n\bar{v}\sigma$. Assuming that the collision cross-section σ is independent of the energy, show that $R_{\text{coll}} \propto \mathcal{N}_{\text{total}}/E_{\text{total}}$ in a harmonic trapping potential. Show that the collision rate increases during evaporation in such a potential.

(10.5) *The properties at the phase transition*

A cloud of 10^6 rubidium atoms is confined in a harmonic trap with oscillation frequencies of $\omega_z/2\pi = 16$ Hz and $\omega_r/2\pi = 250$ Hz (and axial symmetry). Calculate the critical temperature T_C and estimate the density of the cloud at the phase transition.

(10.6) *Properties of a Bose condensate*

The properties of a Bose condensate were calculated in the text using the Thomas–Fermi approximation, which gives accurate results for large condensates. This exercise shows that minimising the energy in eqn 10.33 (a variational calculation) to find the equilibrium size leads to similar results.

In a spherically-symmetric trapping potential, rubidium atoms ($M = 87$ a.m.u.) have an oscillation frequency of $\omega/2\pi = 100$ Hz and hence

$a_{\text{ho}} = 1 \mu\text{m}$. The atoms have a scattering length of $a = 5$ nm. Calculate the following for a condensate with $\mathcal{N}_0 = 10^6$ atoms.

- Show that the repulsive interactions give a much greater contribution to the total energy than the kinetic term.
- Use eqn 10.33 to find an expression for the equilibrium size r and evaluate it.
- What is the density of the condensate?
- Show that the contribution to the energy from the repulsive interactions represents two-fifths of the total.
- Find an expression for the energy E (in terms of $\hbar\omega$). (Note that this expression should have the same dependence on the various parameters as in eqn 10.42, but with a different numerical factor.) Evaluate E/k_B .
- When the trapping potential is switched off suddenly the potential energy goes to zero and the repulsive interaction between the atoms causes the condensate to expand. After a few milliseconds almost all this energy (from the repulsive interactions) is converted into kinetic energy. Estimate the velocity at which the atoms fly outwards and the size of the condensate 30 ms after the trap is switched off.⁴⁹

Comment. These estimates of the important physical parameters show that, although interactions have little influence on the phase transition (Bose–Einstein condensation occurs because of quantum statistics and is completely different to the ‘ordinary’ condensation of a vapour into a liquid caused by molecular interactions, e.g. steam into water), the properties of the condensate itself do depend on the interactions between atoms, e.g. the energy of the condensate is much larger than the zero-point energy of the ground state of the quantum harmonic oscillator.

(10.7) *The chemical potential and mean energy per particle*

- Show that eqn 10.42 follows from the preceding equations in Section 10.6.

⁴⁹The small size of Bose condensates makes them difficult to view directly, although this has been done in certain experiments. Generally, the condensate is released from the trap and allowed to expand before an image (e.g. Fig. 10.12).

- (b) In thermodynamics the chemical potential is the energy required to remove a particle from the system $\mu = \partial E / \partial \mathcal{N}$, where E is the total energy of the system. Show that $E = \frac{5}{7} \mathcal{N} \mu$.

(10.8) *Expansion of a non-interacting condensate*

Although experiments are not carried out with a non-interacting gas it is instructive to consider what happens when $a = 0$. In this case the condensate has the same size as the ground state of a quantum harmonic oscillator (for any $\mathcal{N}_0$) and the initial momentum along each direction can be estimated from the uncertainty principle. For atoms of the same mass as sodium (but with $a = 0$) released from a trap with a radial oscillation frequency of 250 Hz and a frequency of 16 Hz for axial motion, estimate roughly the time of flight at which the cloud is spherical.

(10.9) *Excitations of a Bose condensate*

The vibrational modes of a condensate can be viewed as compression waves that form a standing wave within the condensate; hence these modes have frequencies of the order of the speed of sound divided by the size of the condensate v_s/R . Show that this collective motion of the condensate occurs at a comparable frequency to the oscillation of individual atoms in the magnetic trap.

(10.10) *Derivation of the speed of sound*

The time-dependent Schrödinger equation for the wavefunction of an atom in a Bose–Einstein condensate in a uniform potential is

$$i\hbar \frac{d\psi}{dt} = -\frac{\hbar^2}{2M} \nabla^2 \psi + g |\psi|^2 \psi,$$

where, for simplicity, the potential has been taken as zero ($V = 0$). The wavefunction $\psi = \psi_0 e^{-i\mu t/\hbar}$ satisfies this equation with a chemical potential

$$\mu = g |\psi_0|^2.$$

The trial wavefunction with small fluctuations can be written as

$$\begin{aligned} \psi &= \left[\psi_0 + u e^{i(kx - \omega t)} + v^* e^{-i(kx - \omega t)} \right] e^{-i\mu t/\hbar} \\ &= \psi_0 e^{-i\mu t/\hbar} + \delta\psi(t), \end{aligned}$$

where the amplitudes $|u|$ and $|v|$ are small compared to $|\psi_0|$.

- (a) Show that substituting this function into the Schrödinger equation and making suitable

approximations leads to the same zeroth-order approximation for the chemical potential given above and

$$\begin{aligned} i\hbar \frac{d}{dt} (\delta\psi(t)) \\ = \frac{\hbar^2 k^2}{2M} \delta\psi(t) + g |\psi_0|^2 2\delta\psi(t) + g \psi_0^2 \delta\psi^*(t). \end{aligned} \quad (10.47)$$

- (b) Show that equating terms with the same time dependence leads to two coupled equations for u and v that, in matrix form, are

$$\begin{pmatrix} \epsilon_k + 2g |\psi_0|^2 - \mu & g \psi_0^2 \\ g (\psi_0^*)^2 & \epsilon_k + 2g |\psi_0|^2 - \mu \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \hbar\omega \begin{pmatrix} u \\ -v \end{pmatrix},$$

where $\epsilon_k = \hbar^2 k^2 / 2M$.

- (c) Hence show that u and v are solutions of the matrix equation

$$\begin{pmatrix} \epsilon_k + \mu - \hbar\omega & g \psi_0^2 \\ g (\psi_0^*)^2 & \epsilon_k + \mu + \hbar\omega \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = 0.$$

From the determinant of this matrix, find the relation between the angular frequency of the small oscillations ω and the magnitude of their wavevector k (the dispersion relation). Show that for low energies this gives the same expression for the speed of sound ω/k found in Section 10.7.1.⁵⁰

(10.11) *Attractive interactions*

In certain hyperfine states, the scattering length a of alkali metal atoms changes with the applied magnetic field and this feature has been used to perform experiments in which the atoms have attractive interactions $a < 0$.

Show that eqn 10.33 can be written in the form

$$\frac{4}{3} \frac{E}{\hbar\omega} = x^{-2} + x^2 + Gx^{-3}.$$

By plotting graphs for various values of the parameter G between 0 and -1 , estimate the lowest value of G for which there exists a minimum in the energy as a function of x . For an atomic species with a scattering length of $a = -5$ nm in a trap where $a_{\text{ho}} = 2 \mu\text{m}$, estimate the maximum number of atoms that a Bose condensate can contain without collapsing.

⁵⁰After problem devised by Professor Keith Burnett, Physics graduate class, University of Oxford.